

BIOINFORMATICS MINI-PROJECT 1

Structural Analysis of PDB 1FDL — Fab D1.3–Hen Egg-White Lysozyme Complex

Project overview

This first project introduced me to structural bioinformatics through a real experimentally determined protein complex. I used the RCSB Protein Data Bank and Mol* to inspect PDB 1FDL, identify its protein chains, and focus on the hen egg-white lysozyme component.

1. Research question

What can I learn about an antibody–antigen complex by inspecting its structure, chains and 3D organization?

2. Structure information

Feature	Observation
PDB ID	1FDL
Complex	Fab D1.3–hen egg-white lysozyme
Experimental method	X-ray diffraction
Resolution	2.50 Å
Protein chains	A: antibody light chain; B: antibody heavy chain; C: hen egg-white lysozyme
Lysozyme length	129 residues
Organisms	Mus musculus and Gallus gallus

3. What I did

I opened 1FDL in RCSB PDB, launched Mol*, inspected the chain controls, selected “HEN EGG WHITE LYSOZYME” and Chain C [auth Y], and viewed the complete complex in cartoon representation. A useful practical lesson was that the visible shapes in a molecular viewer should not simply be counted by eye as chains; the chain controls are the reliable way to identify components.



Figure 1. Clean crop of the Mol* structure view showing the 1FDL complex and the selected lysozyme chain. The browser chrome and taskbar have been removed.

4. Biological interpretation

1FDL is an antibody–antigen complex. The lysozyme is the antigen recognized by the D1.3 Fab fragment. The RCSB record describes lysozyme residues 18–27 and 117–125 as part of a discontinuous antigenic determinant involved in antibody contacts, including hydrogen-bond and van der Waals interactions.

5. What I learned

I learned how to move from a PDB record to a 3D molecular structure, identify individual chains, inspect a protein sequence, and connect structural visualization with a biological question.

6. Limitations and next step

This project was mainly observational. I did not calculate contact distances, interface area or structural RMSD. A natural next step is to compare the lysozyme structure quantitatively with another lysozyme structure.

Conclusion

PDB 1FDL provided a useful first exercise because it combines sequence information, multiple protein chains, molecular visualization and a biologically meaningful antibody–antigen interface.